

Class - 01

Why

Statistics is the backbone of data science because it helps us truly understand and interpret data rather than just process it. It allows us to summarize and visualize messy datasets, make reliable decisions using concepts like hypothesis testing and confidence intervals, and select meaningful features through methods like correlation and ANOVA. Most machine learning algorithms are built on statistical foundations, and without stats, it's easy to confuse random noise with real patterns or mistake correlation for causation. Beyond building models, statistics also gives data scientists the language to communicate results clearly and confidently, making it an essential skill for turning raw data into trustworthy insights.

Statistics

Visualization

DATA

So the first question comes to mind is "What exactly is DATA??"

Let's clear something up. Whenever we say *data*, we simply mean **a set of information**.

This word **data** is powerful — it runs the world of Machine Learning, Deep Learning, Data Mining...

Data = Information

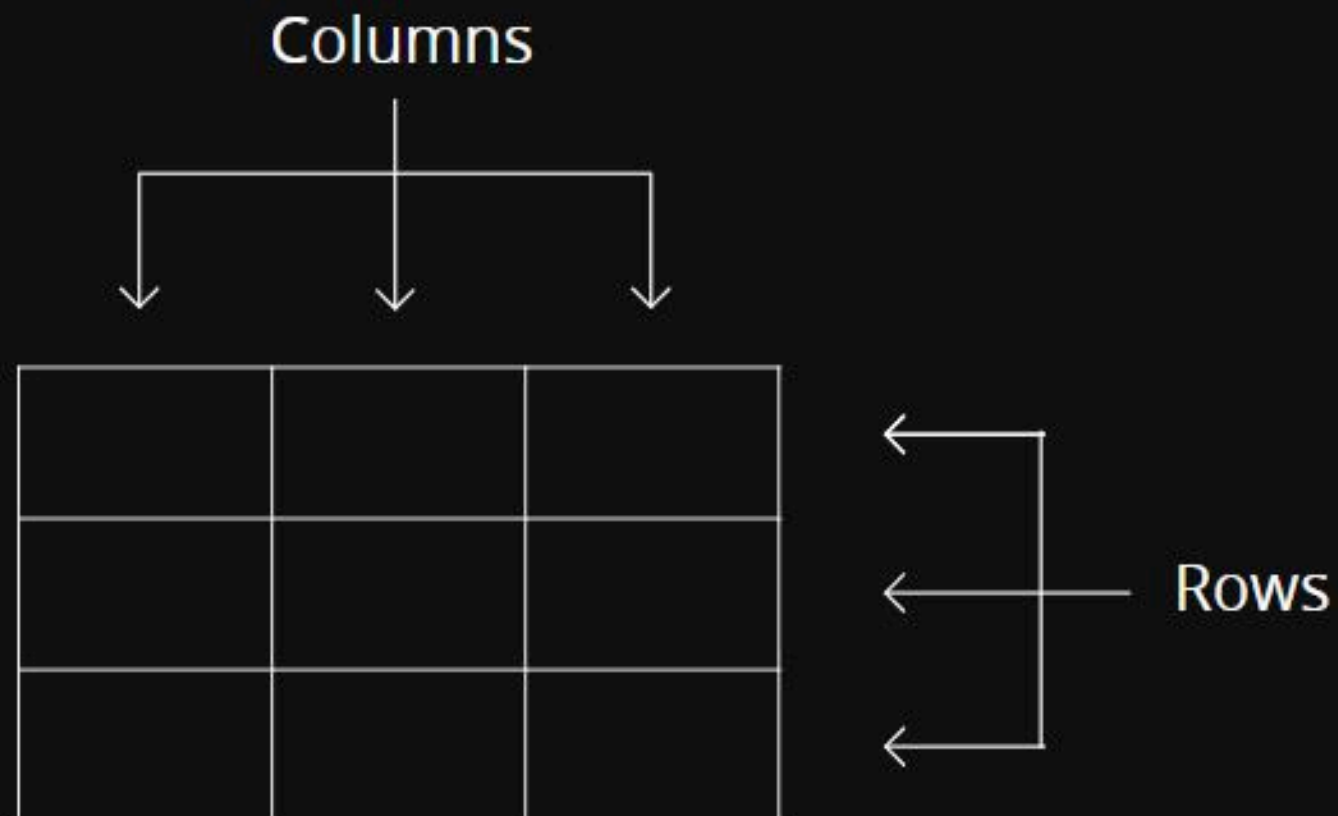
ONE-WAY TABLE

DATA TABLE -

A tabular row & column representation of data

A tabular representation of data is known as a Data Table.

A Data Table has numbers of Rows and Columns in it.



ONE-WAY TABLE

This is what a Simple One-Way Table looks like



In this table there are 2 columns and 4 rows.

Here, Name = Individual

Age = Variable

Name	Age
Vedant	20
Rohan	18
Akarsh	24
Riya	22

ONE-WAY TABLE

Analysing this data we can see Age is represented in numeric values and gender is represented in M and F only.

Continuous numeric data like Age, Salary, Height etc. are known as

Quantitative variables.

Categorical data like gender, Genre etc. are known as **Qualitative variables.**

Individual

Variables



Name	Age	Gender	Place
Vedant	20	M	Bhopal
Rohan	18	M	Delhi
Akarsh	24	M	Indore
Riya	22	F	Raipur

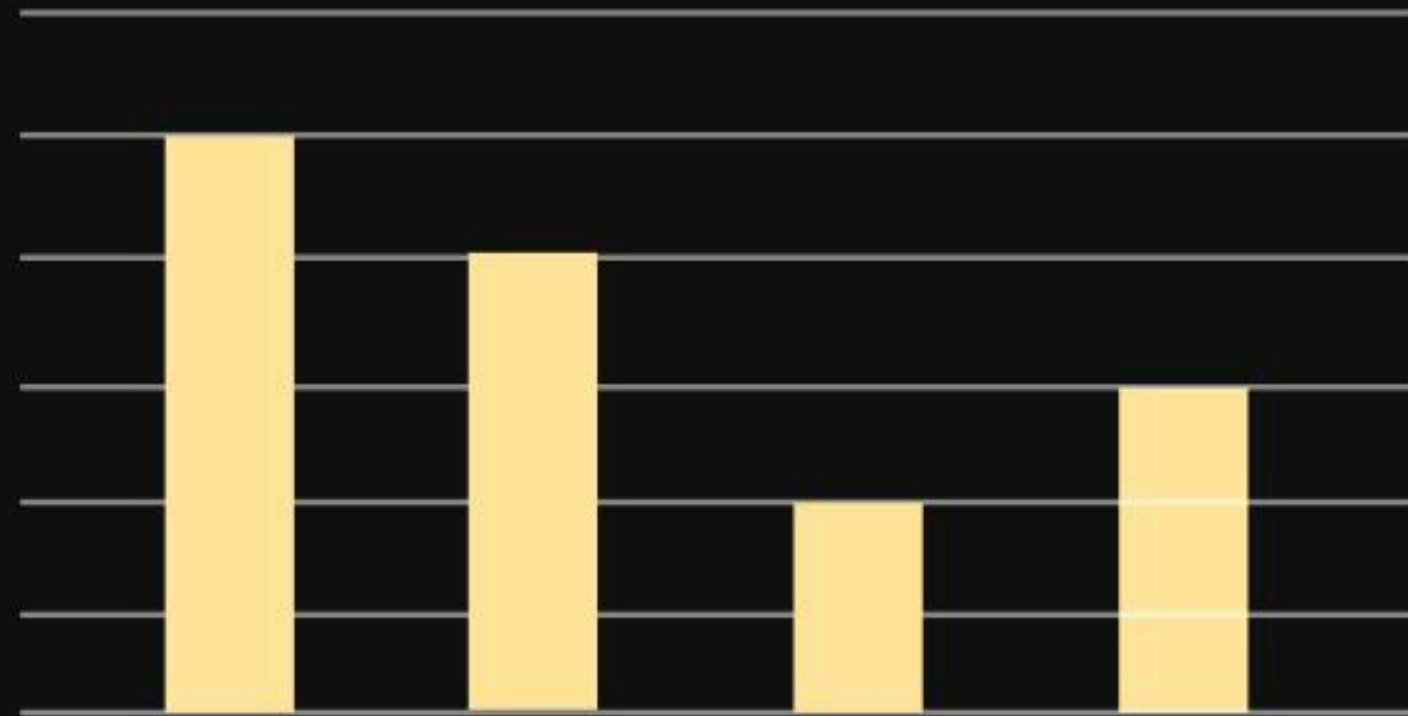
BAR GRAPH

Visualization

BAR GRAPH

So firstly let's talk about " What is Bar Graph ??"

A **bar graph** is a way of showing data using rectangular bars.



BAR GRAPH

This data is just a sample , the real data is ten times bigger than this .

Now the main question comes -

What is the **Highest** Salary?

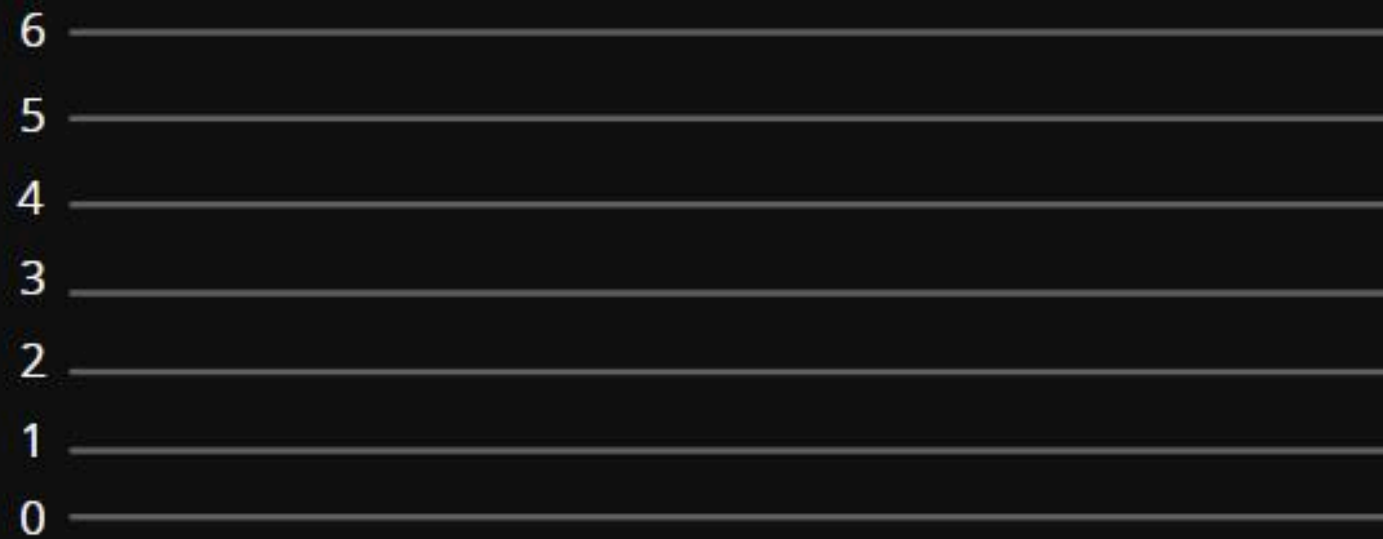
what is the **lowest** Salary?

And to answer these questions we have to count all the highest and lowest salaries can be a headache.

We can answer these question easily if we visualize them using Bar Graph.

Person	Salary
A	1.5
B	2.1
C	1.8
D	2.2
E	1.5
F	1.1
H	1.4
I	2.1
J	1.5
k	1.8

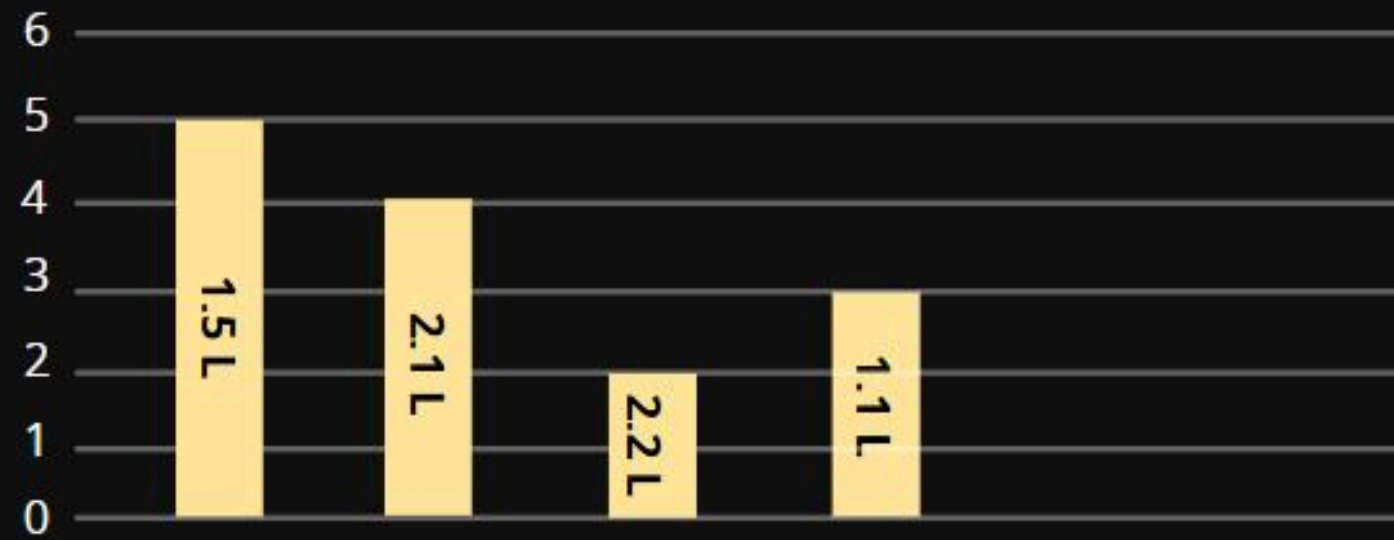
BAR GRAPH



Salary	Count
1.5 L	5
2.1 L	4
2.2 L	1
1.1 L	3

Now to make the just make some horizontal rows and every row has a **value** according to the Salary count.

BAR GRAPH



Salary	Count
1.5 L	5
2.1 L	4
2.2 L	2
1.1 L	3

So there you go we got our **Bar Graph**

PIE CHART

Visualization

PIE CHART

Now Bar graph was good but the next thing we use for visualization is **Pie Chart**.

As bar graph used the frequency counts of fruits.

Pie chart also uses the frequency count of a particular variable.

PIE CHART

But now the question comes " What is a Pie Chart ??"

A pie chart is a ***circle divided into slices***, where each slice shows the proportion of one category out of the total.

But here we have to use the percentage occurrence of every fruits. so lets calculate,

Fruits	Count
Apple	6
Banana	4
Kiwi	2
Mango	2

PIE CHART

Now we need to calculate **percentage** of every count.

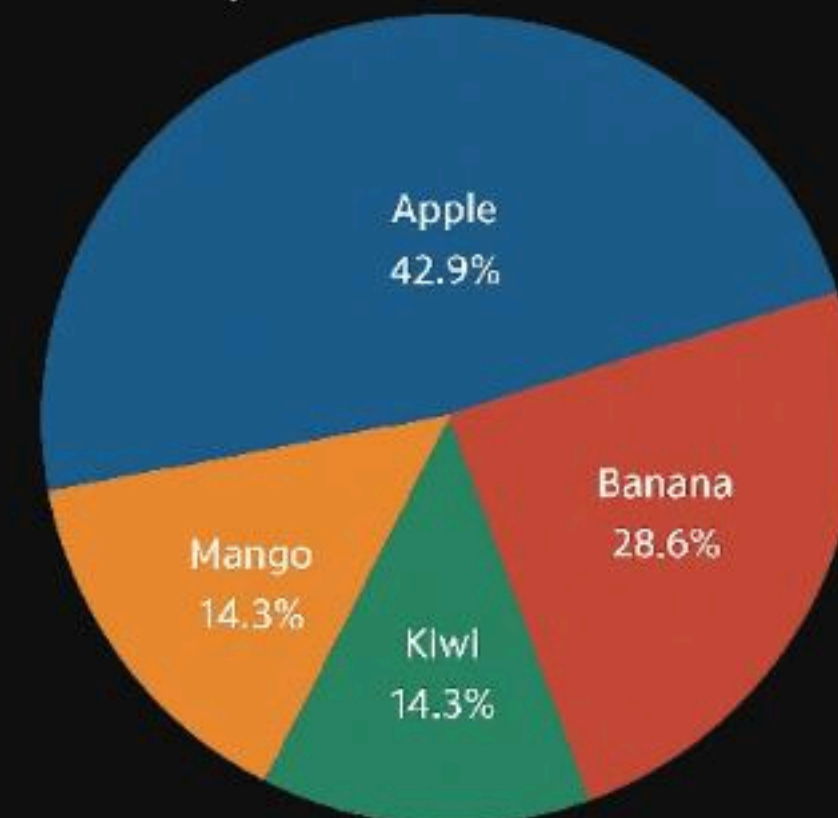
Fruits	Count
Apple	6
Banana	4
Kiwi	2
Mango	2

→ 42.9%

→ 28.6%

→ 14.3%

→ 14.3%



LINE GRAPH

Visualization

Line Graph

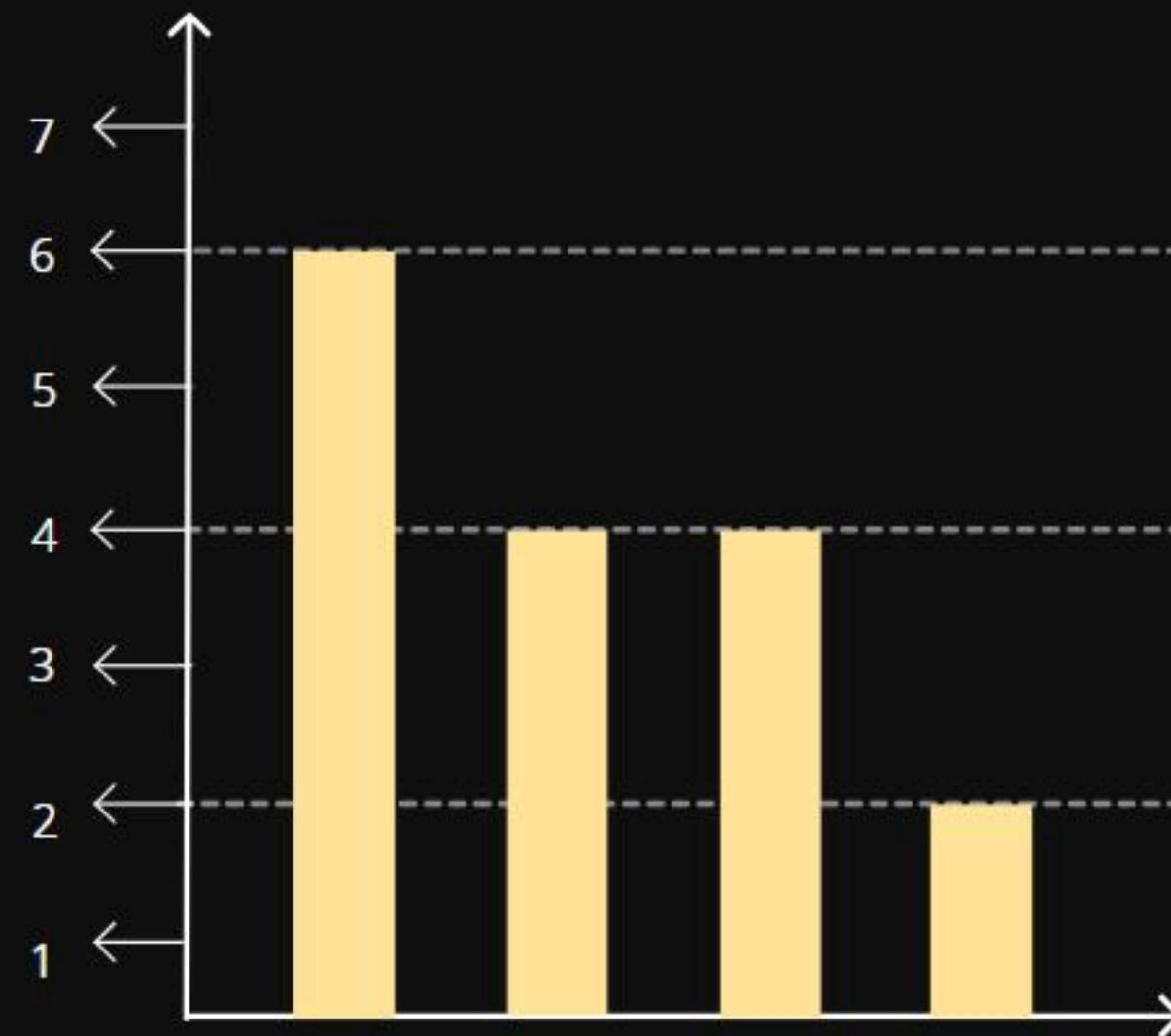
Let me ask you: if I give you a student's marks in **5 tests** and say – “Check whether he/she improved or not”, how would you check?

- A table will give you numbers.
- A bar graph will show the difference.
- But to clearly see the **trend (up or down over time)**, the best tool is a **Line Graph**.

Line Graph

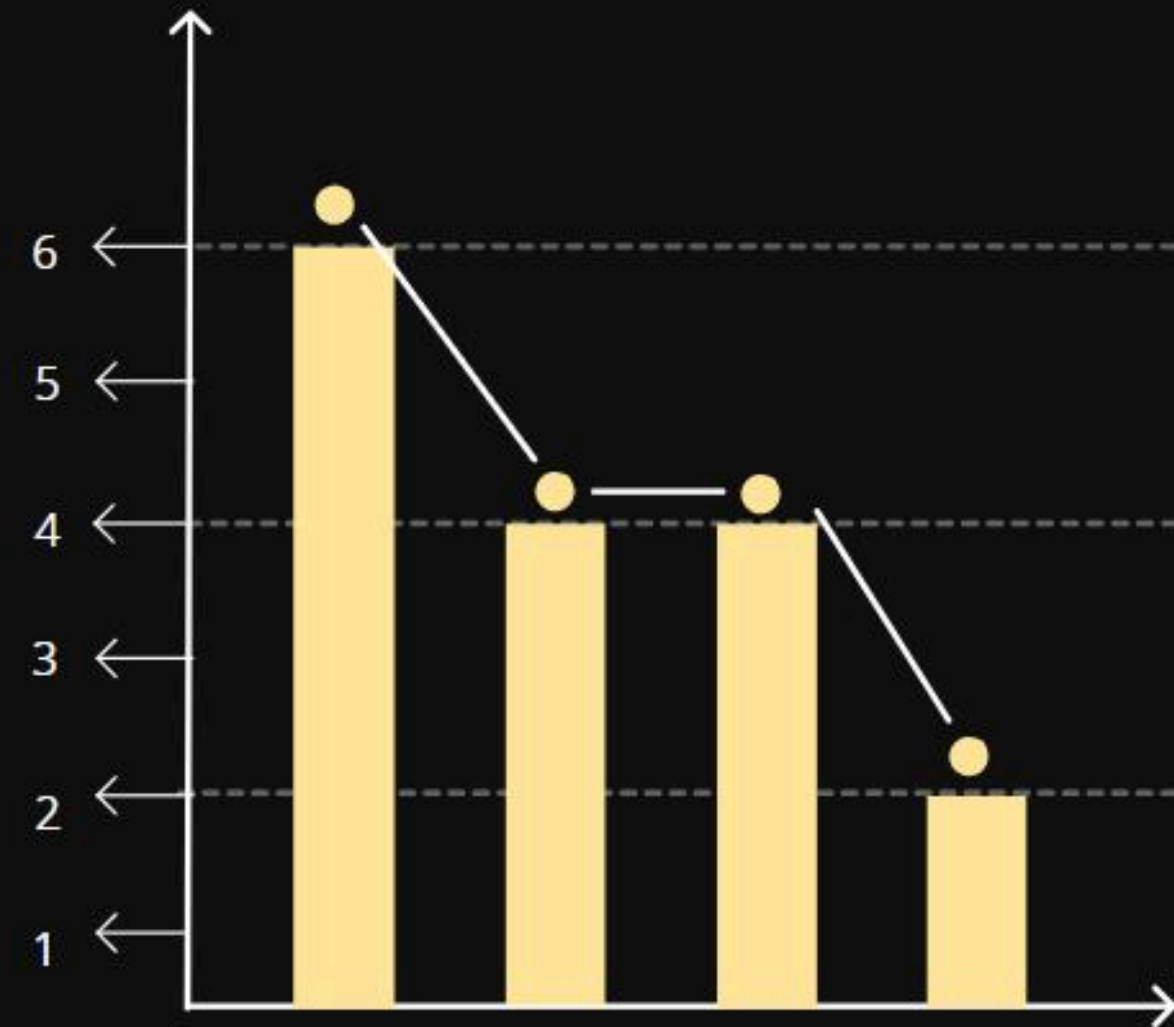
A line graph connects data points with a line, so you can easily see patterns like growth, drop, or stability.

For making a Line Graph you must remember how we made a Bar graph.



Line Graph

Now we put points on each bar and connect them



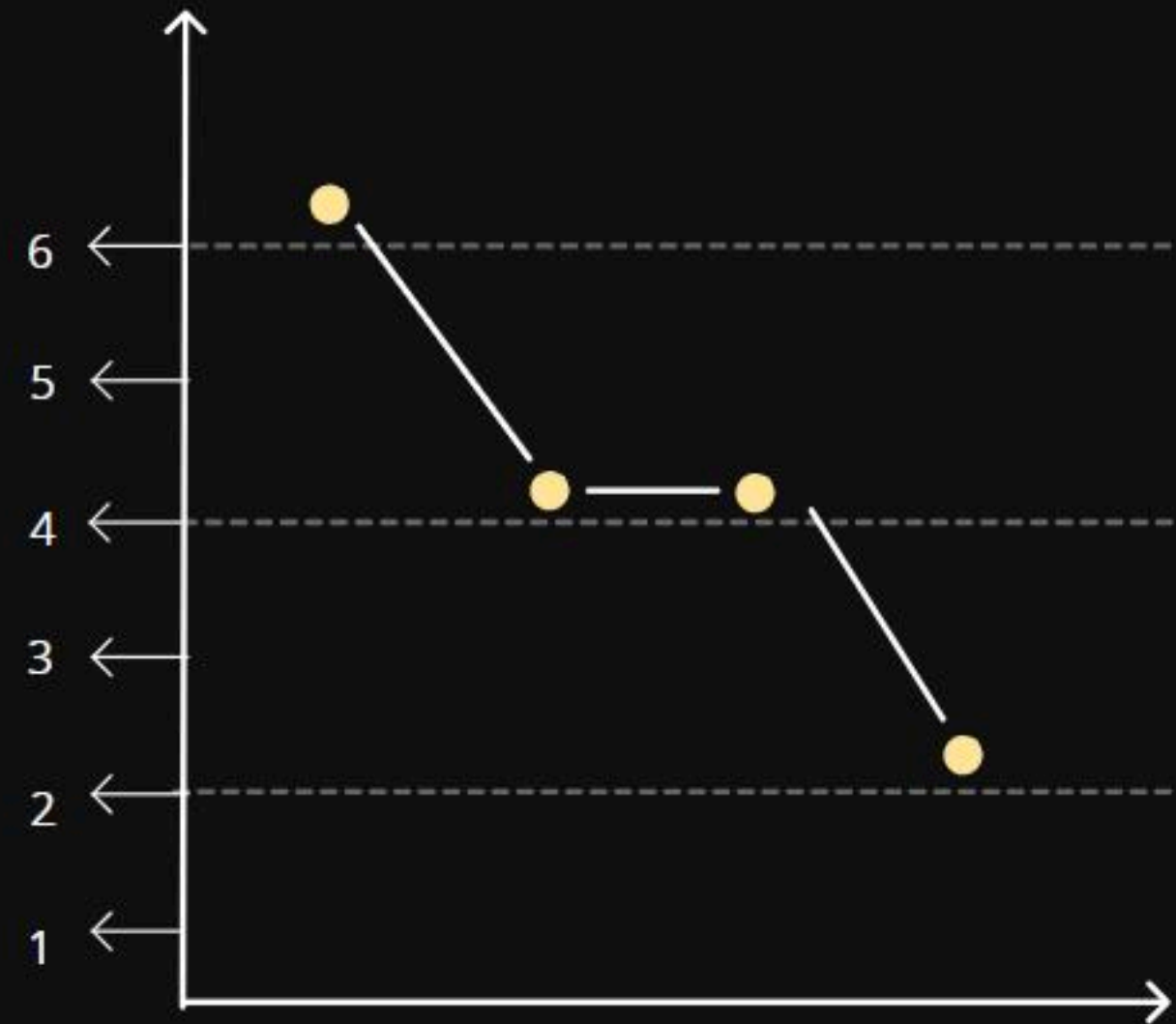
And then we remove the bars

Line Graph

Here we go , we created our Line Graph

But now the question comes When and
How to use the Line Graph?

To answer these questions we will see a data
table.

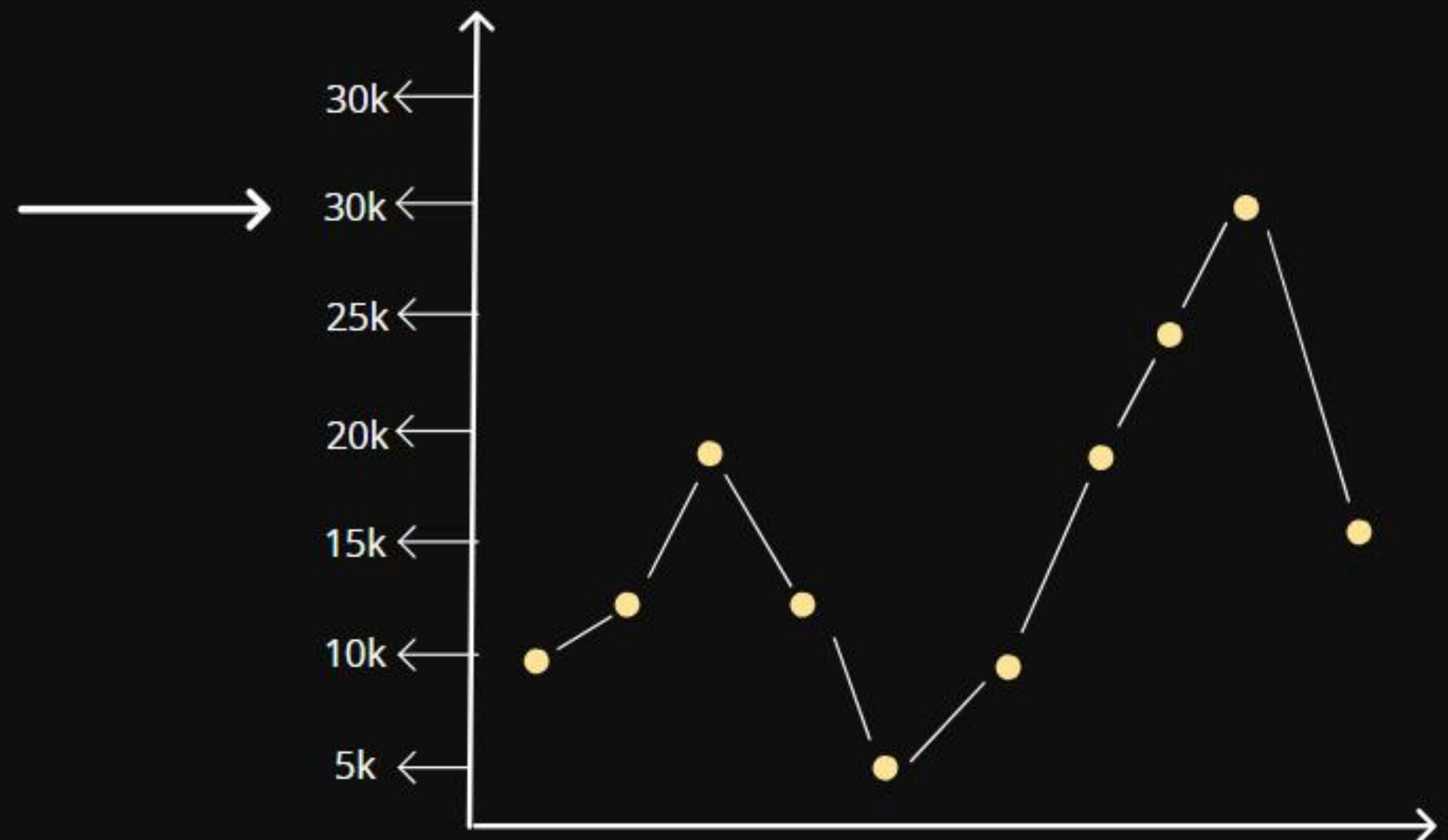


Line Graph

As we can see we have a Monthly expenditure data.

Month	Expenditure
Jan	10k
Feb	12k
Mar	18k
Apr	12k
May	5k
June	10k
July	20k
Aug	25k
Sep	30k
Oct	15k

The Line Graph will show the trends of our continuous data.



O-GIVES

Visualization

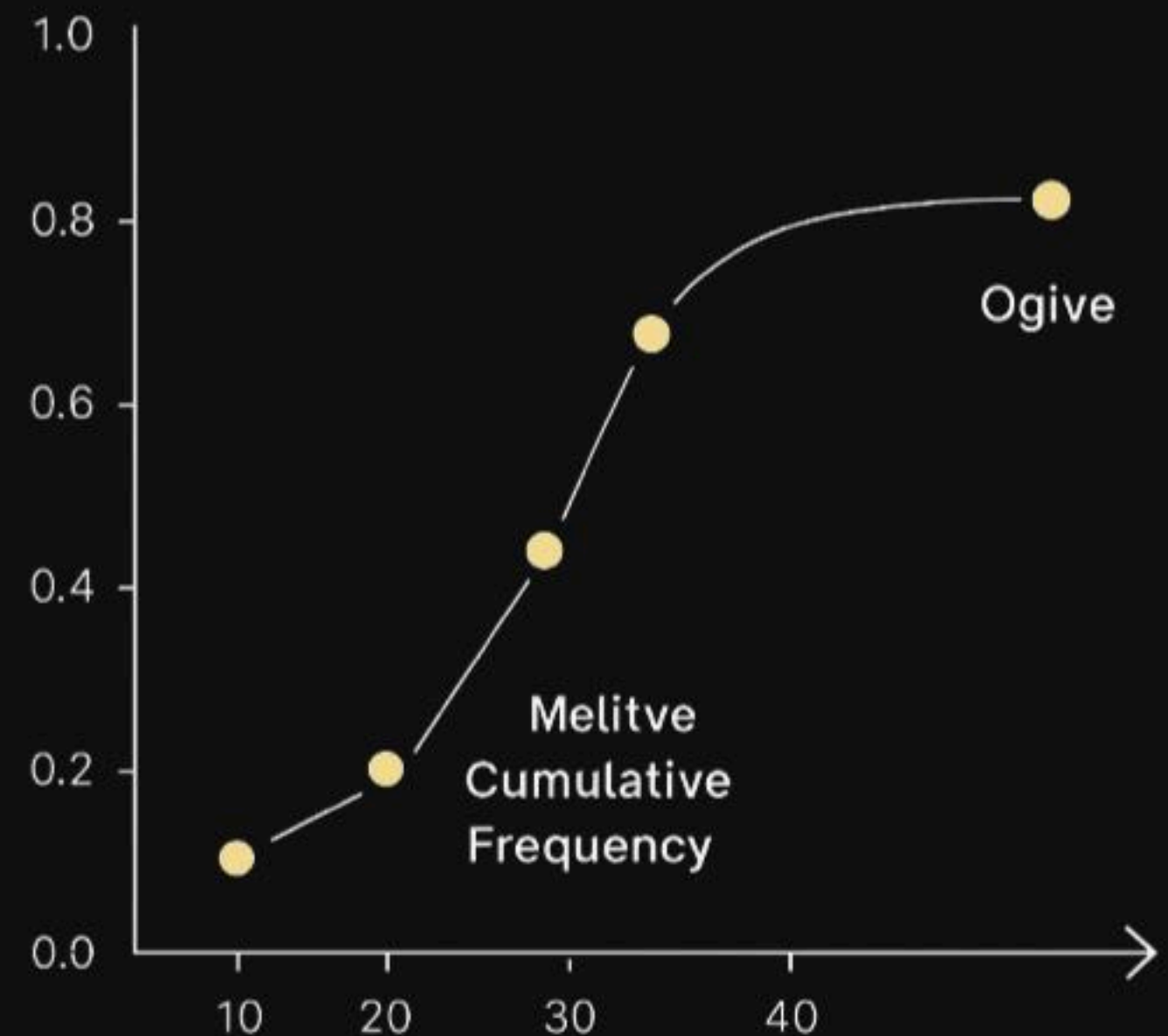
O-Gives

What's an O-Gives ??

An O-Gives is a graph of **Cumulative Frequency**.

It tells you :

- “how many values are **up to** a point” (less-than O-Give)
- or “how many values are **at least** a point” (more-than O-Give).



O-Gives

Month	Expenditure
Jan	5k
Feb	10k
Mar	12k
Apr	7k
May	25k
June	20k
July	18k
Aug	25k
Sep	15k
Oct	15k



Month	Expenditure
Jan	5k
Feb	15k
Mar	27k
Apr	34k
May	59k
June	79k
July	97k
Aug	122k
Sep	137k
Oct	152k

) +
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O-Gives

O-Gives

```
graph TD; OGives[O-Gives] --- LTOG[Less - than O-Give]; OGives --- MTOG[More - than O-Give]
```

Less - than O-Give

In this Cumulative Frequency is **Plotted up to the upper class limit.**

The Curve goes **upwards** Left to Right.

More - than O-Give

In this Cumulative Frequency is **Plotted from the lower class limit.**

The Curve goes downwards.

TWO-WAY TABLE

Visualization

Two-Way Table

A **two-way table** shows the relationship between **two variables at the same time**.

Let's try to understand the two way table with an example,

Here's the data of customers visiting a cafe in 2022,

Here in this table Month is individual (independent) where as Customers are a Dependent variable as revenue depends on month.

Month	Customers
Jan	200
Feb	250
Mar	300
Apr	180
May	400
June	350

Two-Way Table

Let's suppose that the data is of 3 years

But this One-way table can only gives us the data of several months or a single year and we don't have the data of every year.

Month	Customers
Jan	200
Feb	250
Mar	300
Apr	180
May	400
June	350

Two-Way Table

Month	Customers
Jan	200
Feb	250
Mar	300
Apr	180
May	400
June	350

So now we broke down the data for every year and made it into table and this table data is known as a **Two-Way Table**.



Month	2022	2023	2024
Jan	180	200	220
Feb	230	250	270
Mar	280	300	320
Apr	160	180	190
May	370	400	420
June	340	350	360

DOT PLOT

Visualization

Dot Plot

Now let's understand about Dot Plots.

Score	Frequency
5	1
6	6
7	3
8	5
9	2
10	4

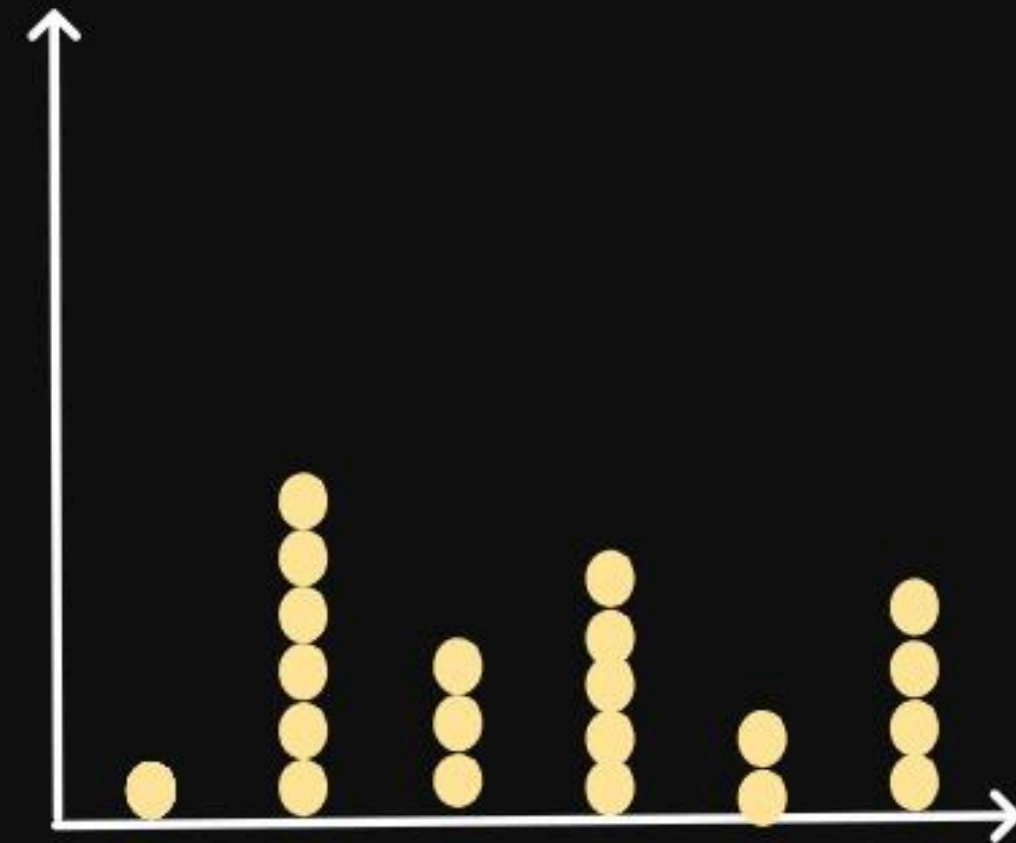
For creating a Dot Plot we have to use the frequency table of specific data which fulfil the need.

Let's say we have the test scores of 10 students in a class -

Dot Plot

"A Dot Plot is like a Bar Graph but instead of Bars we use **Dots Stacking on top of each other.**"

Score	Frequency
5	1
6	6
7	3
8	5
9	2
10	4



And here we go we got our Dot Plot

Relative Frequency Table

Visualization

Relative Frequency Table

Let's start by understanding the Relative Frequency Table -

A Relative Frequency Table shows how often each value occurs in **proportion to the total**, usually written as a **Fraction, Decimal, or Percentage**.

In simple words, Instead of just counts, it tells us “**what fraction / percentage of the total**” each category represents.

Let's try to understand this with an example

Relative Frequency Table

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Let's try to understand this with an example

Relative Frequency Table

let's say we have students who prefer different modes of transport to college like *Bus* and *Bicycle*

Two-Way Table



Students	Bus	Scooty
Boys	30	10
Girls	20	40

Now if we only focus on Bus we can capture many things like but one of the thing is **Percentages**.

Lets calculate the percentage of Bus and Scooty.

Relative Frequency Table

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Lets calculate the percentage of Bus and Scooty.

Relative Frequency Table

The formula for Relative Frequency is :

$$\text{Relative Frequency} = \frac{\text{Frequency}}{\text{Total}} \times 100$$

Now if we only focus Column wise, we can calculate percentages.

Students	Bus	Scooty
Boys	30	10
Girls	20	40

Relative Frequency Table

Students	Bus	Scooty
Boys	30	10
Girls	20	40



So after Calculating the percentage
Column wise.

Students	Bus	Scooty
Boys	0.60 / 60%	0.10 / 10%
Girls	0.40 / 40%	0.80 / 80%

↓
100%

↓
100%

Similarly we can also create a **Row related frequency table.**

JOINT DISTRIBUTION

Visualization

Joint Distribution

A joint Distribution table is similar to a relative frequency table as it also contains percentages.

But in a Joint Distribution we will **compare two different distributions together**.

It tells us: “ What is the probability of both X and Y happening together? ”

Joint Distribution

Let's tackle Joint Distribution with an example

		Travelling Mode		
Student Category	Students	Bus	Scooty	Total
	Boys	30	10	40
	Girls	20	40	60
	Total	50	50	100

Now we'll find the Percentage

Joint Distribution

Let's tackle Joint Distribution with an example

		Travelling Mode		
Student Category	Students	Bus	Scooty	Total
	Boys	30	10	40
	Girls	20	40	60
	Total	50	50	100

Now we'll find the Percentage

Joint Distribution

		Travelling Mode		
Student Category	Students	Bus	Scooty	Total
	Boys	30%	10%	40%
	Girls	20%	40%	60%
	Total	50%	50%	100%

There you go this is a joint distribution as there is a **Distribution for travelling mode** and **Distribution of Student** and we are comparing both the distribution.

HISTOGRAM

Visualization

Histogram

A histogram is a type of bar graph that is used to show the **distribution of numerical data**.

- Instead of categories (like in bar graphs), histograms deal with **ranges of numbers** called **intervals or bins**.
- It helps us see **how data is spread out** : is it concentrated in one place, spread evenly, or skewed?
- The **height of each bar** shows how many data points fall inside that interval.

In simple words -

In Histogram we group **continuous data** into **buckets / bins** and then counting how many items fall into each bucket.

Histogram

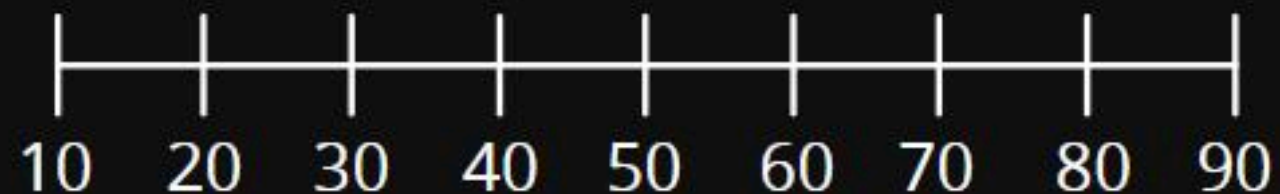
Candidate	Age
P1	21
P2	22
P3	45
P4	56
—	—

Assume we have a data of candidates and their ages.

But just looking at raw numbers doesn't tell us much about the overall pattern.

To understand the distribution, we use a **histogram**.

Now to make a histogram we have to again use a X axis, and on that axis we can create the bins portrayed below.

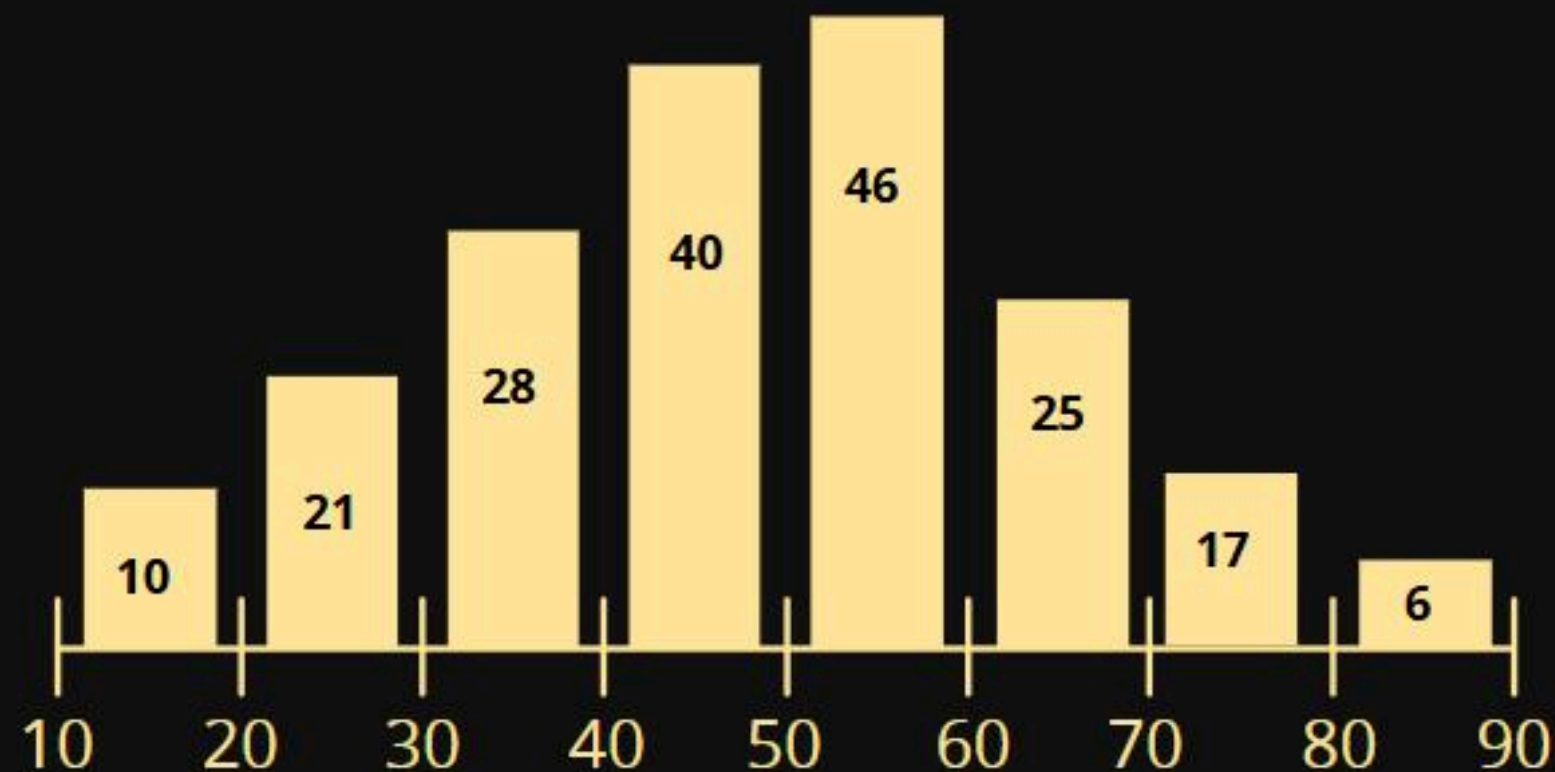


Histogram

Now inside these bins we have to put the candidates according to there age.

Its kind of capturing the frequency of ages like, how many times specific portion of ages occurred.

After this you have to plot the bars according to the number of candidates in bins.



Now we got our Histogram